

Specific Probability Distributions

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Bernoulli trials : If a random experiment results in only two outcomes is known as Bernoulli trial. The outcome in which one is interested is known as success and the other one is known as failure.

EX. 1) Toss a coin, outcomes are H, T
If one is interested in H, it is success and tail T is failure.

11) Throw a die : Even no. or odd no. makes it Bernoulli trial

Bernoulli Prob Distⁿ: (Discrete prob distⁿ) If a discrete random variable X takes only two values 1 with prob. p and 0 with prob $q = (1-p)$ ($0 < p < 1$, $0 < q < 1$), then the random variable X is said to follow a Bernoulli prob distⁿ. Its prob mass function $p(x)$ is written as

$$P(X=x) = p(x) = p^x (1-p)^{1-x}; x=0,1$$
$$= p^x q^{1-x}; x=0,1$$

$$X \sim B(1, p)$$

$$q = 1-p.$$

In tabular form

X	$p(x)$
0	q
1	p
1.00	

Toss a fair coin

X	$p(x)$
0	$\frac{1}{2}$
1	$\frac{1}{2}$
1.00	

$$X(H) = 1$$
$$X(T) = 0$$

$$\text{Mean } (\mu_1') = E(X) = 0 \cdot (1-p) + 1 \cdot p = p$$

$$\mu_2' = E(X^2) = 0^2 \cdot (1-p) + 1^2 p = p$$

$$\text{variance } \mu_2 = \mu_2' - \mu_1'^2 = p - p^2 = p(1-p) = pq.$$

Binomial Prob distribution :

It is a discrete prob distⁿ defined for discrete r.v.
generalization of bernoulli prob distⁿ.

Let there be a sequence of n (finite) independent & identical bernoulli trials such that in each trial prob of success p is constant, hence the prob of failure $q = (1-p)$ in each trial is also constant. The prob of x no. of success in n such trials will be obtained as

$$A_1 : SSFSSFFSSFFFSFSSFSS \dots FSS \quad ; \quad \begin{array}{l} x \text{ no. of } S + (n-x) \\ \text{no. of } F \end{array}$$

$$\text{Hence } P(A_1) = P(S)P(S)P(F) \dots P(F)P(S)P(S) \quad (\text{trials are independent})$$

$$= p^x q^{n-x}$$

Similarly

$$A_2 : SFFSFSFSF \dots SFFS$$

$$P(A_2) = p^x q^{n-x}$$

x no. of success out of n such trials can be obtained in $\binom{n}{x}$ ways

...

$$A_{\binom{n}{x}} : SFFSSSFFSFSF \dots FFSFSF$$

$$P(A_{\binom{n}{x}}) = p^x q^{n-x}$$

Events $A_1, A_2, \dots, A_{\binom{n}{x}}$ are m. excl/m.d. events.

$$\text{Hence } P(X=x) = p(x) = P(A_1 \cup A_2 \cup \dots \cup A_{\binom{n}{x}})$$

$$\boxed{p(x) = \binom{n}{x} p^x q^{n-x} ; x = 0, 1, 2, \dots, n}$$

Defⁿ:

A discrete r.v. X is said to follow a binomial prob distⁿ with parameters n (finite no. of ^{indep} bernoulli trials) and p (prob of success, $0 < p < 1$) if its prob mass function is given as

$$P(X=x) = p(x) = \binom{n}{x} p^x q^{n-x} ; x = 0, 1, 2, \dots, n$$

$$q = 1-p \quad (\text{prob. of failure})$$

Remark (1) $X \sim B(n, p)$

ii) $p(x)$ is pmf.

It is obvious that $p(x) > 0 \forall x = 0, 1, 2, \dots, n$ and

$$\sum_{x=0}^n p(x) = \sum_{x=0}^n \binom{n}{x} p^x q^{n-x} = (q+p)^n = 1$$

Hence, $p(x)$ is pmf.

iii) frequency function $f(x) = N p(x)$

It indicates the no. of times we will get x no. of success if an experiment consists n independent bernoulli trials repeated N times.

$f(x)$ is the frequency nearest integer will be considered its value.

$$\text{Ex } f(4) = 100 \times 0.48375 = 48.375 \approx 48$$

Here $x=4$, $N=100$, $p(4)=0.48375$.

Moment Generating Function of $B(n, p)$

$$X \sim B(n, p), \quad p(x) = \binom{n}{x} p^x q^{n-x}; \quad x=0, 1, 2, \dots, n.$$

$$M_X(t) = E e^{tx} = \sum_{x=0}^n e^{tx} p(x) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x q^{n-x}$$

$$= \sum_{x=0}^n \binom{n}{x} (pe^t)^x q^{n-x} = (q + pe^t)^n$$

$$\boxed{M_X(t) = (q + pe^t)^n}$$

$$\text{Mean } \mu_1' = \left. \frac{\partial M_X(t)}{\partial t} \right|_{t=0}; \quad \frac{\partial M_X(t)}{\partial t} = n(q + pe^t)^{n-1} pe^t$$

$$\text{Put } t=0, \text{ we get } \left. \frac{\partial M_X(t)}{\partial t} \right|_{t=0} = np$$

$$\text{Hence, } \boxed{\text{Mean} = np}$$

$$\mu_2' = \left. \frac{\partial^2 M_X(t)}{\partial t^2} \right|_{t=0}; \quad \frac{\partial^2 M_X(t)}{\partial t^2} = n(n-1)(q + pe^t)^{n-2} (pe^t)^2 + n(q + pe^t)^{n-1} pe^t$$

$$\text{Hence } \mu_2' = \left. \frac{\partial^2 M_X(t)}{\partial t^2} \right|_{t=0} = n(n-1)p^2 + np$$

$$\therefore \text{variance } \mu_2 = \mu_2' - \mu_1'^2 = n(n-1)p^2 + np - n^2p^2 = np(1-p) = npq \text{ (variance)}$$

$$\boxed{\text{Mean} > \text{Variance}} \quad \text{as } 0 < p < 1 \text{ and } 0 < q < 1.$$

Additive Property of $B(n, p)$

$$X_1 \sim B(n_1, p_1) ; M_{X_1}(t) = (q_1 + p_1 e^t)^{n_1} ; (q_1 = 1 - p_1)$$

$$X_2 \sim B(n_2, p_2) ; M_{X_2}(t) = (q_2 + p_2 e^t)^{n_2} ; (q_2 = 1 - p_2)$$

$$M_{X_1+X_2}(t) = M_{X_1}(t) M_{X_2}(t)$$

$$= (q_1 + p_1 e^t)^{n_1} (q_2 + p_2 e^t)^{n_2}$$

$$\neq (q + p e^t)^n \text{ (can't be written in m.g.f form of } B(n, p))$$

Hence, in general it does not hold the additive property.

But, as particular case; if $p_1 = p_2 = p$, $q_1 = q_2 = q$

We will have

$$M_{X_1+X_2}(t) = (q + p e^t)^{n_1} (q + p e^t)^{n_2}$$

$$= (q + p e^t)^{n_1+n_2} \text{ which is the m.g.f of}$$

$B(n_1+n_2, p)$, follows additive property as particular case.

Hence if $X_1 \sim B(n_1, p)$ + $X_2 \sim B(n_2, p)$

$$X_1 + X_2 \sim B(n_1 + n_2, p)$$

Ex $X_1 \sim B(7, \frac{2}{3}), X_2 \sim B(5, \frac{2}{3})$

$$X_1 + X_2 \sim B(12, \frac{2}{3})$$